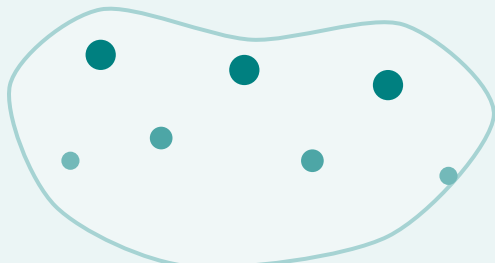


Reference Geometry

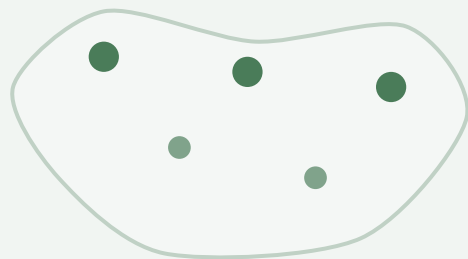
HLCA (Healthy)



$$z_H \in \mathbb{R}^{30}$$

$$T_\phi : d_H \mapsto d_L$$

LuCA (Cancer)

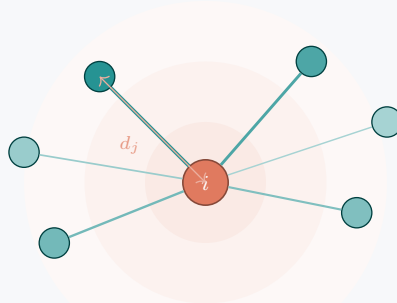


$$z_L \in \mathbb{R}^{10}$$

$$T_\phi^* = \arg \min_T \sum_{ij} |d_H^{ij} - d_L^{T(i), T(j)}|^2$$

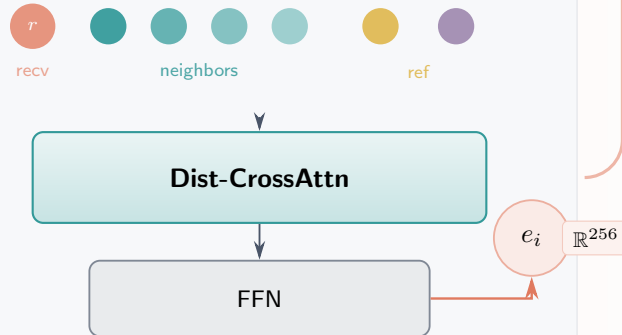
Niche Encoding

Spatial Niche



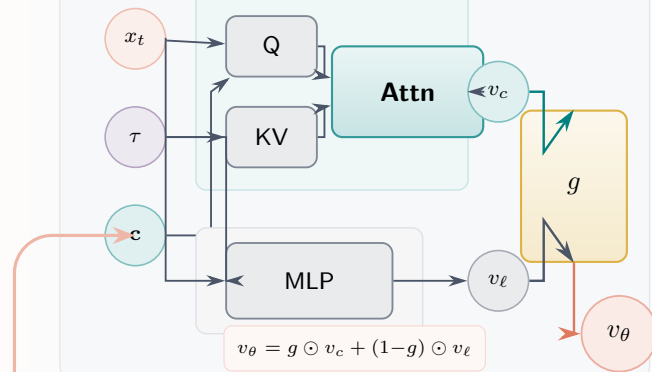
$$\alpha_j \propto e^{-d_j/\sigma_d} \cdot e^{q^T k_j/\sqrt{d}}$$

Niche Encoder

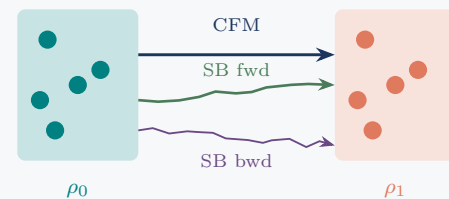


Transport & Drift

CrossAttentionDrift



OT-CFM / Schrödinger Bridge



$$\mathcal{L}_{\text{cfm}} = \|v_\theta - (x_1 - x_0)\|^2$$

$$\mathcal{L}_{\text{sb}} = \|s_\psi - \nabla \log p_t\|^2$$

Mathematical Formulation

Core Equations

1. Distance-Weighted Attention (AMICI)

$$\alpha_{ij} = \frac{\exp(-d_{ij}/\sigma_d) \cdot \exp(q_i^\top k_j / \sqrt{d})}{\sum_{k \in \mathcal{N}_i} \exp(-d_{ik}/\sigma_d) \cdot \exp(q_i^\top k_k / \sqrt{d})}$$

Spatial prior decays with distance; modulates standard scaled dot-product attention.

2. Niche Context Embedding

$$e_i = \text{CrossAttn}(Q = r_i, KV = [\mathcal{N}_i, z_H, z_L, p]) \in \mathbb{R}^{256}$$

Receiver queries local neighbors, dual-atlas latents, and 14 PROGENy pathway activities.

3. Gromov-Wasserstein Atlas Fusion

$$T_\phi^* = \arg \min_T \sum_{i,j} |d_H(x_i, x_j) - d_L(T_i, T_j)|^2$$

$$z_f = [z_H \parallel T_\phi(z_H)] \in \mathbb{R}^{40}$$

4. CrossAttentionDrift (Gated Velocity)

$$v_\theta(x_t, t \mid \mathbf{c}) = g \odot v_c + (1 - g) \odot v_\ell, \quad g = \sigma(W_g[x_t \parallel \tau \parallel \mathbf{c}])$$

Transport Dynamics

OT-CFM (Conditional Flow Matching)

$$x_t = (1 - t)x_0 + tx_1, \quad (x_0, x_1) \sim \pi_{\text{OT}}^*$$

$$\mathcal{L}_{\text{cfm}} = \mathbb{E}_t [\|v_\theta(x_t, t \mid \mathbf{c}) - (x_1 - x_0)\|^2]$$

Schrödinger Bridge

$$\text{Fwd: } dx_t = v_\theta(x_t, t \mid \mathbf{c}) dt + \sigma dW_t$$

$$\text{Bwd: } dx_t = (v_\theta - \sigma^2 s_{t,t}) dt + \sigma d\bar{W}_t$$

Symbol Reference

Symbol	Dim	Description
x_0, x_1	$\mathbb{R}^{40}$	Source/target cell states
d_{ij}	$\mathbb{R}^+$	Receiver-neighbor distance
$\mathcal{N}_i$	set	Spatial neighborhood of cell i
z_H, z_L	30, 10	HLCA/LuCA atlas latents
p	$\mathbb{R}^{14}$	PROGENy pathway activities
$e_i, \mathbf{c}$	$\mathbb{R}^{256}$	Niche context embedding
v_θ, s_ψ	$\mathbb{R}^{40}$	Velocity / Score network